

Algorithms for Computer Detection of Symmetry Elements in Molecular Systems

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Simple procedures for the location of proper and improper rotations and reflexion planes are presented. The search is performed with a molecule divided into subsets of symmetrically equivalent atoms (SEA) which are analyzed separately as if they were a single molecule. This approach is advantageous in many aspects. For instance, in those molecules that are symmetric rotors, the number of atoms and the inertia tensor of the SEA provide one straight way to find proper rotations of any order. The algorithms are invariant to the molecular orientation and their computational cost is low, because the main

information required to find symmetry elements is interatomic distances and the principal moments of the SEA. For example, our Fortran implementation, running on a single processor, took only a few seconds to locate all 120 symmetry operations of the large and highly symmetrical fullerene C_{720} , belonging to the I_h point group. Finally, we show how the interatomic distances matrix of a slightly unsymmetrical molecule is used to symmetrize its geometry. © 2013 Wiley Periodicals, Inc.

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Introduction

In many molecules, after their building atoms are rotated around an axis or reflected through a mirror plane, the resulting molecule will be completely indistinguishable from the initial one. These axes and planes are called symmetry elements and the movements described are symmetry operations. Because the set of all existing symmetry operations for a given molecule, together with the identity operation, forms a mathematical group, the Group Theory can be used to make qualitative predictions of many physical properties. For instance, a molecule with two or more axes of symmetry that are not collinear, does not have a dipole moment because the dipole is a vector quantity and cannot be moved by any symmetry operation. Group theory is extensively used in molecular spectroscopy, where useful information like selection rules, degeneracies and perturbations due to resonance can be determined.^[1–4] It is also widely used in electronic structure programs, for example, to classify states and orbitals or to save computational resources in several steps of the calculations.^[5–11]

While locating symmetry elements in molecules is in general a simple task for humans, who can either use molecular models or visualization software packages, a computer code must do the same starting from a set of coordinates and masses only. Many electronic structure programs possess routines to find the molecule point group without need for any further assistance from the user,^[12–16] but the procedures followed by them to locate symmetry elements were not listed in the literature. However, methods to determine the molecular symmetry using Graph Theory were reported by Ivanov and Schüürmann^[17,18] and Balasubramanian.^[19,20] Conversely, Avnir et al. treat symmetry as a continuous property where the location of symmetry elements is transformed into a problem of minimization of a specific function related to the corresponding symmetry operation.^[21–23] With these methods, the CPU

time can become limiting for large and highly symmetrical systems, for example, fullerenes, because the calculation time is a factorial function of the number of atoms. Approximations to circumvent the scaling problem are available for cyclic symmetry point groups^[24] and for structures with a high topological symmetry.^[18]

In this article, we present a set of algorithms to locate the vectors corresponding to point group symmetry elements from the coordinates of a set of punctual masses. Some methods to find symmetry elements^[25] test their existence along or perpendicular to the principal axes of inertia or examine the axes connecting the center of mass and one atom or the midpoint of like-atoms. In the current approach, the molecule is divided in sets of symmetrically equivalent atoms^[3] (SEA) which are analyzed separately. The information obtained from one set of SEA is then compared to that obtained from the other ones and only those symmetry operations common to all sets can exist in the entire molecule and need to be tested. Dividing the molecule into sets of SEA is advantageous when searching for symmetry elements because often the number of SEA is lower than the number of atoms in the molecule. In addition, when the molecule is a symmetric or an asymmetric rotor, the number of SEA and its inertia tensor greatly simplify the location of all existing rotations. Such operations are by

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far the most difficult to find, inasmuch as they can be of any order. The algorithms are invariant to the molecular orientation. The main required information is obtained from the interatomic distance matrix, tensors of inertia or directly from the atomic coordinates. They were implemented in our program PLACZEK^[26–28] to enable the automatic exploitation of the point group symmetry in infrared and Raman spectra calculations.

In the next section, we describe the procedure to locate symmetry elements followed by the “Numerical Examples” section, which shows in detail how the point group of ammonia and allene can be found from their Cartesian coordinates and masses. Several examples of outputs from PLACZEK showing the location of symmetry elements are given as Supporting Information.

Methodology

The location of symmetry elements starts with the classification of the molecule in one of the rotor types: (i) linear, (ii) spherical, (iii) symmetric or (iv) asymmetric. This information is particularly useful for linear systems, where all symmetry elements are collinear or orthogonal to the principal axes of inertia, and also to deal with spherical top molecules as explained later. After the rotational symmetry is determined, the atoms are divided into groups of symmetrically equivalent ones and the inertia tensor of each set is calculated. Then the number of SEA and their respective principal moments and axes are used to find the symmetry elements as detailed in the next sections.

Symmetrically equivalent atoms

Most symmetrical molecules have identical atoms that are interchanged by symmetry operations, as the hydrogens of water by C_2 rotation. Such atoms are called^[3]SEA and their presence is readily recognized by analyzing the interatomic distance matrix, $\mathbf{D}$. If a molecule has k SEA then there will be k columns in $\mathbf{D}$ with identical values and these columns will correspond to atoms of the same isotopic type. For instance, the interatomic distance matrix of water is:

$$\mathbf{D} = \begin{array}{ccc|c} \text{H}_1 & \text{O} & \text{H}_2 & \\ \hline & 0 & d_{\text{OH}} & d_{\text{HH}} \\ d_{\text{OH}} & & 0 & d_{\text{OH}} \\ d_{\text{HH}} & d_{\text{OH}} & & 0 \end{array} \begin{array}{l} \text{H}_1 \\ \text{O} \\ \text{H}_2 \end{array}$$

The equal distances listed in columns 1 and 3 show that water hydrogens are symmetrically equivalent.

Locating symmetry elements

In the following sections, the methodology used for the location of proper and improper rotations and reflexion planes is presented. This methodology is completely independent of the molecule orientation but requires the center of mass to be

in the origin because it cannot be moved by any symmetry operation.

Proper rotation: C_n . The search for the proper rotations is performed separately for each set of SEA through the analysis of the principal moments of inertia of the set, I_A^{SEA} , I_B^{SEA} , and I_C^{SEA} . The existing rotations in a given set depend on the type of arrangement formed by its atoms that will be one of the following: (i) single atom, (ii) linear, (iii) polygonal and, (iv) polyhedral arrangement. The distinction among these possibilities is easily made by inspecting I_A^{SEA} , I_B^{SEA} , and I_C^{SEA} moments.

Single Atom. The single atom set itself gives no information about the existence of symmetry elements in the molecule but as this atom is not moved by any symmetry operation, its coordinates can be useful to locate symmetry elements (axes and planes). The oxygen of water and the nitrogen in nitrobenzene are examples of this case.

Linear Arrangement. This arrangement occurs when the SEA set has two atoms and is characterized by the inertia moments of the linear rotor ($I_A^{\text{SEA}} = 0$ and $I_B^{\text{SEA}} = I_C^{\text{SEA}}$). Any order rotations along the SEA axis can exist for the entire molecule as well as C_2 rotations perpendicular to this axis, passing in the center of mass of the SEA. Examples of this arrangement are the hydrogens of water and the oxygens in carbon dioxide.

Polygonal Arrangement. When the principal moments of the SEA match the condition of the planar arrangement ($I_A^{\text{SEA}} + I_B^{\text{SEA}} = I_C^{\text{SEA}}$), these atoms will form a k -sided polygon of the type:

- Regular: if the moments of inertia are those of the oblate symmetric top ($I_A^{\text{SEA}} = I_B^{\text{SEA}} \neq I_C^{\text{SEA}}$). This arrangement is exemplified by the carbons in cyclopropane.
- Irregular: when the SEA are arranged as in an asymmetric rotor ($I_A^{\text{SEA}} \neq I_B^{\text{SEA}} \neq I_C^{\text{SEA}}$). The oxygens of *p*-dinitrobenzene illustrate this situation (Fig. 1).

The polygonal SEA can have rotation axes perpendicular to the polygon plane passing through its center of gravity. In the regular polygon the possible C_n are those with $n = k$ and the divisors of k , but only k divisors can occur in irregular polygons. In cyclopropane $k_C = 3$ and $I_A^{\text{SEA}} = I_B^{\text{SEA}} \neq I_C^{\text{SEA}}$, thus this SEA set possesses a C_3 perpendicular to the carbons' plane. In allene, where $k_H = 4$ and $I_A^{\text{SEA}} \neq I_B^{\text{SEA}} \neq I_C^{\text{SEA}}$, the unique possibility is the C_2 .

Polyhedral Arrangement. Another spatial disposition for the SEA is illustrated in Figure 2 where the symmetrically equivalent carbons of ferrocene form a prism. Depending on the atoms' mass and the distance between base and top polygons, the prismatic arrangement will be identified either by the principal moments of the prolate symmetric top ($I_A^{\text{SEA}} < I_B^{\text{SEA}} = I_C^{\text{SEA}}$) or those of the oblate symmetric top ($I_A^{\text{SEA}} = I_B^{\text{SEA}} < I_C^{\text{SEA}}$). The highest order rotation possible is the $C_{k/2}$ along the axis passing through the center of mass of the polygons and the C_2 perpendicular to this axis can also be present. In the example of Figure 2, where $k_C = 10$, the existing rotations are the C_5 and the perpendicular C_2 . Polyhedral SEA may also be spatially

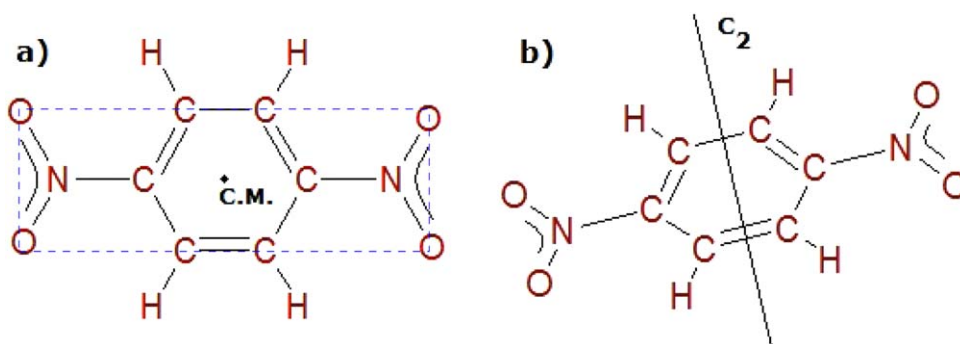


Figure 1. The oxygens in *p*-dinitrobenzene are symmetrically equivalent and form an irregular polygon (a) with one C_2 axis perpendicular to the atoms' plane (b). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

arranged as a primitive orthorhombic unit cell. This case is characterized by the asymmetric rotor moments ($I_A^{\text{SEA}} \neq I_B^{\text{SEA}} \neq I_C^{\text{SEA}}$) and presents three C_2 along the principal axes of inertia.

Proper Rotation of Order two: C_2 . For a nonlinear molecule with no more than two SEA, the highest order rotation possible is the C_2 . In this case, the I_A^{SEA} , I_B^{SEA} , and I_C^{SEA} axes themselves are insufficient to locate the C_2 because the only information they give is that an existing C_2 must be perpendicular to the I_C^{SEA} axis, but there is an infinitude of axes satisfying this condition. Moreover, a molecule belonging to one of the dihedral groups D_n , D_{nh} , and D_{nd} possesses n C_2 rotations perpendicular to the highest order C_n that cannot be located only by the principal moments of the SEA. To find all the C_2 axes, with the direction vector being $\mathbf{r}_{C_2}$, one or more of the following procedures need to be used:

- When the C_2 cross the middle point of two atoms in the SEA set and this middle point is not the center of mass of the molecule (e.g., the hydrogens in water or neighbor carbons in benzene):
- The positions of any two atoms from the SEA set are taken and their middle point is calculated, $\bar{\mathbf{r}}_{ij} = (\mathbf{r}_i + \mathbf{r}_j)/2$.

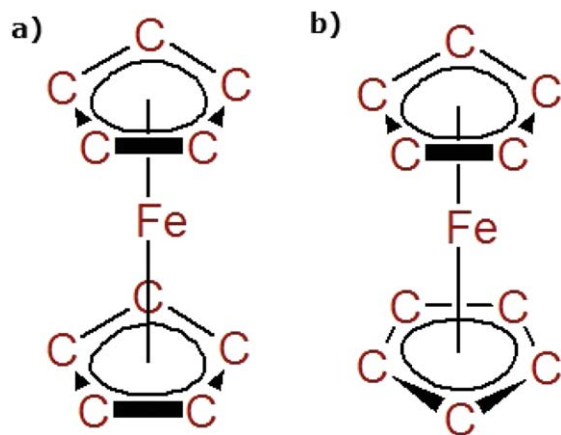


Figure 2. Carbons in eclipsed (a) and anti- (b) conformations of ferrocene are examples of SEA arranged as prism and antiprism.

- If $\bar{\mathbf{r}}_{ij} \neq \mathbf{r}_{\text{CM}}$ then another vector is constructed: $\mathbf{r}_{C_2} = \bar{\mathbf{r}}_{ij} - \mathbf{r}_{\text{CM}}$. Vector $\mathbf{r}_{\text{CM}}$ is the molecule center of mass position.

- Vector $\mathbf{r}_{C_2}$ can pass through one or more atoms of the SEA (e.g., two of the four C_2 axes in $\text{C}_4\text{O}_4^{2-}$ anion cross a pair of atoms in the SEA):

- In this case, $\mathbf{r}_{C_2} = \mathbf{r}_i - \mathbf{r}_{\text{CM}}$, with $\mathbf{r}_i$ being the position of one (any) atom in the SEA set.

- For molecules with two or more sets of diatomic SEA (linear arrangement), an existing C_2 must be simultaneously perpendicular to $\mathbf{r}_{ij}$ and $\mathbf{r}_{kl}$ as defined below:

- first, one vector connecting the two SEA is built: $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. Analogously, a second vector $\mathbf{r}_{kl}$ is constructed using the positions of two atoms from other diatomic SEA.
- From the orthogonality condition, we have $\mathbf{r}_{C_2} = \mathbf{r}_{ij} \times \mathbf{r}_{kl}$.

An example of the need for procedure **C** is in the location of C_2 in trans-1,2-dichloroethene where its six atoms form three sets of diatomic SEA.

Note that some of the $\mathbf{r}_{C_2}$ created following the procedures above may not be valid C_2 vectors (e.g., $\mathbf{r}_{C_2}$ generated from carbons of ferrocene following **A** or **B**). Therefore, it is necessary to check if the rotation around $\mathbf{r}_{C_2}$ is indeed a valid C_2 in the molecule. For dihedral groups, an additional step may be necessary where a dot product between $\mathbf{r}_{C_2}$ and previously found $\mathbf{r}_{C_n}$ is used to confirm that this C_2 is perpendicular to the highest order rotation, C_n .

Locating the Highest Order Proper Rotation, C_n . The complete procedure to identify the possible rotations in each SEA set is summarized in the algorithm of Figure 3. After identifying C_n , their existence must be checked in the entire molecule. Only the rotations shared by all sets can be present in the molecule. Thus, if a given set possesses both C_4 and C_2 but another has only the C_2 , the highest allowed rotation for the entire molecule will be the C_2 .

The reason for this additional verification is that if two different SEA sets possess a C_n of the same n there is no

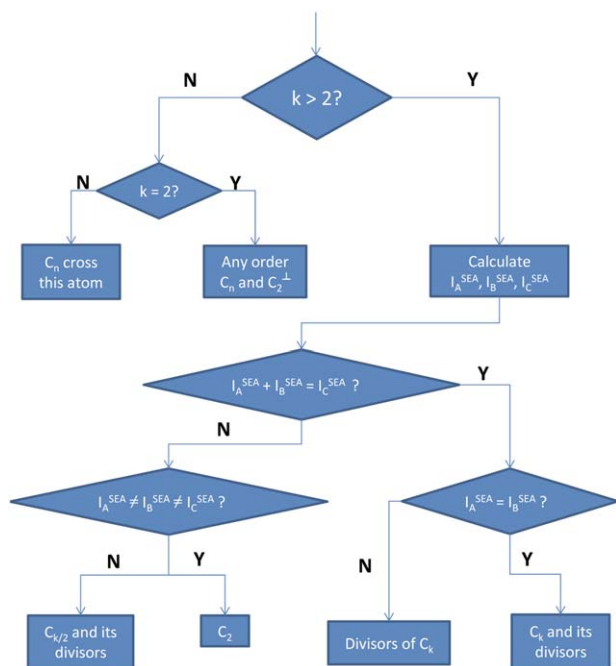


Figure 3. Algorithm used to locate proper rotations in sets of SEA based on the analysis of their number, k , and principal moments of inertia, I_A^{SEA} , I_B^{SEA} , and I_C^{SEA} . [Color figure can be viewed in the online issue, which is available at [wileyonlinelibrary.com](http://onlinelibrary.wiley.com).]

guarantee that the two rotations are the same. This uncertainty exists because each SEA set is analyzed separately into a new frame where its inertia tensor is evaluated as if the SEA were a single molecule. Only linear molecules do not need to be checked because their point group is determined from the molecular inertia tensor and inversion operation.

For polygonal SEA, the proper rotation is perpendicular to the atoms' plane and cross the center of the polygon. This axis is given by the eigenvector of the principal moment of inertia I_C^{SEA} . In the polyhedral arrangement, the C_n axis will be the eigenvector corresponding to the I_A^{SEA} inertia moment for SEA of the prolate top symmetry or I_C^{SEA} for the oblate top symmetry.

Spherical Arrangement. Molecules that belong to the point groups of highly symmetrical objects (tetrahedrons, octahedrons, and icosahedrons) are easy to identify because their principal moments of inertia are triply degenerated. For such molecules, the inertia tensors of the SEA sets are not as useful as in symmetric and asymmetric rotors to locate proper rotations. The current approach to find rotations in spherical rotors explores the information of their point groups that limits the number and type of each kind of C_n (for instance, if a molecule has icosahedric symmetry, it will have $15C_2$, $20C_3$ and $24C_5$ axes). The point group is identified as being of the T , O or I type from the number of C_2 found following procedures **A** and **B** presented in section "Proper rotation of order two." Once this number is determined the quantity of C_3 , C_4 , and C_5 will be known, making the search for C_n in spherical molecules more efficient.

Proper rotations of order three can be found using the procedure:

- The positions of any three atoms in the SEA set are taken to build vectors $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$, $\mathbf{r}_{ik}$ and $\mathbf{r}_{jk}$.
- If $\mathbf{r}_{ij}$, $\mathbf{r}_{ik}$ and $\mathbf{r}_{jk}$ have the same norm then atoms A_i , A_j and A_k form an equilateral triangle thus enabling the possibility of the molecule to have a C_3 axis perpendicular to these atoms' plane.
- The C_3 operation will be checked by rotating the molecule around vector $\mathbf{r}_{C_3} = \mathbf{r}_{ij} \times \mathbf{r}_{jk}$.

When the number of detected C_3 reaches their quantity in the corresponding point group the procedure is interrupted. The high order rotations C_4 and C_5 are located following an analogous procedure.

Improper rotation axes: S_{2n} and S_n . Whenever the improper rotation needs to be verified, the S_{2n} or S_n axes will be collinear to the already known C_n axis. Therefore, improper rotations are checked simply by applying these operations to the molecule.

Reflexion planes: σ . The procedure for finding symmetry planes divides the SEA in subgroups of two atoms where the coordinates of the atomic pairs are used to generate a vector normal to the plane, $\mathbf{n}$. Supposing atom B is the mirror image of A , then the coordinates of these atoms can be written in the form: $\mathbf{r}_A = \mathbf{r}_A^{\parallel} + \mathbf{r}_A^{\perp}$ and $\mathbf{r}_B = \mathbf{r}_A^{\parallel} - \mathbf{r}_A^{\perp}$. The vector $\mathbf{r}_A^{\parallel}$ is the projection of $\mathbf{r}_A$ into the mirror plane and $\mathbf{r}_A^{\perp}$ the projection orthogonal to the plane. Therefore, the vector $\mathbf{n}$ is obtained from the difference $\mathbf{r}_A - \mathbf{r}_B$. When the symmetry plane is one of the Cartesian planes, the pair $A-B$ is easily recognized. Otherwise the following procedure is adopted:

1. A SEA set containing two or more atoms is chosen and any one of its atoms is assumed to be the A atom.
2. Any other atom from this set is taken to be B and a candidate of $\mathbf{n}$ is constructed together with the matrix of the symmetry operation σ .
3. When the current $\mathbf{n}$ is indeed the normal vector, after operation of σ the only observed effect is the interchange of SEA.
4. If $\mathbf{n}$ is not the normal vector then steps 2 and 3 are repeated choosing another atom as B . The maximum number of attempts in a SEA set of k atoms is $k - 1$, usually a very small number.
5. In the case of no planes found another set of SEA is selected for searching.

Molecules with no SEA must be either planar or linear to possess reflexion planes. Both situations are readily identifiable from the molecular principal moments of inertia. An additional step in the above procedure may be required in order to classify the symmetry plane as σ_v , σ_d or σ_h but this is a simple task when $\mathbf{n}$ vector and the highest order C_n axis are already known.

Inversion. The existence of inversion is checked by inspecting the Cartesian coordinates in a frame where the center of mass is in the origin. When inversion is present, the coordinates of

Table 1. Possible point groups for each type of rotor.

Rotor type	Point groups
Linear	$C_{\infty v}$, $D_{\infty h}$
Spherical Top	T_d , T_h , O_h , I_h , K
Symmetric Top	C_n , C_{nh} , C_{nv} , D_n , D_{nh} , D_{nd} , S_n
Asymmetric Top	C_1 , C_s , C_2 , C_{2v} , C_{nh} , C_{nv} , D_n , D_{nh} , D_{nd} , S_n

the atom i , $\mathbf{r}_i = (x_i, y_i, z_i)$, will appear in another atom of the same isotopic type with their signs changed, $\mathbf{r}_j = -\mathbf{r}_i$. The only exception occurs if that atom is located in the origin (if any).

Classification of molecules into point groups

It is well known that a small number of symmetry elements are sufficient to assign the molecule to its point group.^[4] In addition, the principal moments of inertia I_A , I_B , and I_C are useful to simplify this search because each type of rotor can belong uniquely to certain point groups as shown in Table 1.^[3] Our computer implementation for finding the point group, first classifies the molecule according to its rotor type: (i) linear, $I_A = 0$ and $I_B = I_C$, (ii) spherical, $I_A = I_B = I_C$, (iii) symmetric, $I_A = I_B \neq I_C$ or $I_A \neq I_B = I_C$, and (iv) asymmetric, $I_A \neq I_B \neq I_C$. The next steps will depend on the rotor type:

- Linear rotor: In this simple case, if the molecule possesses a center of inversion, its point group is $D_{\infty h}$, and otherwise is $C_{\infty v}$.
- Spherical rotor: the number of C_2 rotations and reflexion planes and the existence (or nonexistence) of inversion is sufficient to classify the molecule as T , T_d , T_h , O , O_h and I_h .
- Symmetric and asymmetric rotors: the point group is ascertained based on presence/absence of some key symmetry elements (e.g., see Figs. 1–33 of Ref. [4]).

Examples of computing time used by our Fortran implementation are shown in Table 2. These represent examples obtained using a single processor, to find all symmetry elements and their operations for some large molecules. For the largest molecule, fullerene C_{720} , which belongs to the I_h point group, only a few seconds were needed to locate its 120 symmetry operations.

Nearly symmetrical molecules

For many reasons, for example, numerical imprecision, we sometimes deal with unsymmetrical molecules with a geome-

try that does not deviate significantly from the perfectly symmetrical structure. In such situations, if more symmetrical structure is needed, the current approach offers one way to perform a geometry symmetrization calculation. The algorithm presented here, divides the problem into two parts: (1) Symmetrization of the interatomic distances matrix and (2) finding a new set of atomic coordinates corresponding to the symmetrized matrix.

Average interatomic distance matrix: $\langle D \rangle$. The interatomic distance matrix D is symmetrized using the following steps:

1. A threshold δ_D to consider zero the difference between pairs of interatomic distances, $|D_{ij} - D_{kl}|$, is set to some value. Then the routine that groups atoms into sets of SEA is called and the number of SEA sets found, N^{SEA} , is registered. This procedure is repeated many times, adding small increments in δ_D , seeking the δ_D that produces the lowest value of N^{SEA} . We name this value δ_D^{min} . The idea here is that a small number of SEA sets can be associated with high symmetry (e.g., Buckminsterfullerene has a single SEA set, $N^{SEA} = 1$, with 60 atoms).
2. We already know that two SEA have a one-to-one correspondence between elements of their columns in D . In a perfectly symmetrical molecule there are N pairs satisfying $|D_{ij}^{SEA} - D_{kl}^{SEA}| = 0$, where N is the number of atoms in the molecule and D_{ij}^{SEA} , D_{kl}^{SEA} are one of the N matching pairs. Conversely, in a slightly unsymmetrical molecule, $|D_{ij}^{SEA} - D_{kl}^{SEA}| > 0$, however, $|D_{ij}^{SEA} - D_{kl}^{SEA}| < \delta_D^{min}$. Therefore, in the next step we perform the symmetrization of D in order to have all N pairs $|D_{ij}^{SEA} - D_{kl}^{SEA}| = 0$. The algorithm for symmetrization will be described using an example. Suppose a slightly unsymmetrical molecule of ammonia, with the following interatomic distances:

$$D = \begin{bmatrix} 0 & d_{H_1H_2} & d_{H_1H_3} & d_{H_1N} \\ d_{H_2H_1} & 0 & d_{H_2H_3} & d_{H_2N} \\ d_{H_3H_1} & d_{H_3H_2} & 0 & d_{H_3N} \\ d_{NH_1} & d_{NH_2} & d_{NH_3} & 0 \end{bmatrix}$$

where

$$d_{H_1H_2} > d_{H_1H_3} > d_{H_2H_3}$$

$$d_{NH_1} > d_{NH_2} > d_{NH_3}$$

$$|d_{H_1H_2} - d_{H_1H_3}| > |d_{H_1H_3} - d_{H_2H_3}|$$

Table 2. Computing time, evaluated using the *gprof* tool, required by our Fortran code to find all symmetry elements and the point group of some molecules.

Molecule	Chemical Formula	N_{atom}	Point Group	Time/s
Triphenylborane	$B(C_6H_5)_3$	34	D_3	0.02
Buckminsterfullerene	C_{60}	60	I_h	0.11
Hexapyridineiron (II)	$Fe(C_5H_5N)_6^{2+}$	67	T_h	0.02
Octaphenyloctasilsesquioxane	$Si_8O_{12}(C_6H_5)_8$	108	D_4	0.03
Fullerene C_{720}	C_{720}	720	I_h	3.17

If $(d_{H_1H_2} - d_{H_2H_3}) < \delta_D^{\min}$ and $(d_{NH_1} - d_{NH_3}) < \delta_D^{\min}$, then hydrogens will be grouped into one single set just as it occurs in perfectly symmetrical ammonia. Starting with distance $d_{H_2H_1}$ appearing in the column corresponding to hydrogen H_1 , the closest counterpart from the column of H_2 is $d_{H_2H_1}$ (indeed identical, since $\mathbf{D}$ is symmetrical) and from the H_3 column is $d_{H_1H_3}$. Now, these three distances are replaced by their average,

$$\begin{aligned}\langle d_{H_2H_1} \rangle &= \frac{d_{H_2H_1} + d_{H_1H_2} + d_{H_1H_3}}{3} \\ \langle d_{H_2H_1} \rangle &= \langle d_{H_2H_1} \rangle \\ \langle d_{H_1H_3} \rangle &= \langle d_{H_2H_1} \rangle\end{aligned}$$

which are stored in matrix $\langle \mathbf{D} \rangle$. Similarly,

$$\begin{aligned}\langle d_{H_3H_1} \rangle &= \frac{d_{H_3H_1} + d_{H_3H_2} + d_{H_2H_3}}{3} \\ \langle d_{H_3H_2} \rangle &= \langle d_{H_3H_1} \rangle \\ \langle d_{H_2H_3} \rangle &= \langle d_{H_3H_1} \rangle \\ \langle d_{NH_1} \rangle &= \frac{d_{NH_1} + d_{NH_2} + d_{NH_3}}{3} \\ \langle d_{NH_2} \rangle &= \langle d_{NH_1} \rangle \\ \langle d_{NH_3} \rangle &= \langle d_{NH_1} \rangle\end{aligned}$$

Because $d_{H_1H_2} \neq d_{H_2H_3}$, the averages $\langle d_{H_2H_1} \rangle \neq \langle d_{H_3H_1} \rangle$, thus, the next step is to compute another average using these elements:

$$\begin{aligned}\langle \overline{d_{H_2H_1}} \rangle &= \frac{\langle d_{H_2H_1} \rangle + \langle d_{H_3H_1} \rangle}{2} \\ \langle \overline{d_{H_2H_1}} \rangle &= \frac{\langle d_{H_1H_2} \rangle + \langle d_{H_1H_3} \rangle + \langle d_{H_2H_3} \rangle}{3} \\ \langle \overline{d_{H_3H_1}} \rangle &= \langle \overline{d_{H_2H_1}} \rangle\end{aligned}$$

Moving to nitrogen, we have:

$$\begin{aligned}\langle d_{H_1N} \rangle &= \frac{d_{H_1N} + d_{H_2N} + d_{H_3N}}{3} \\ \langle d_{H_2N} \rangle &= \langle d_{H_1N} \rangle \\ \langle d_{H_3N} \rangle &= \langle d_{H_1N} \rangle\end{aligned}$$

Finally we conclude that $\langle d_{H_iN} \rangle = \langle d_{NH_i} \rangle$, with $i = 1, 2$ or 3 , because $\mathbf{D}$ is symmetric, and, therefore, the new matrix $\langle \mathbf{D} \rangle$ is also symmetric.

Symmetrized Cartesian coordinates: $\mathbf{r}_i^{\text{sym}}$

Now we need to determine a set of coordinates $\mathbf{r}_1 \cdots \mathbf{r}_N$ with the interatomic distances matrix $\langle \mathbf{D} \rangle$. These coordinates can be found minimizing the quadratic deviation function:

$$S(\mathbf{r}_1 \cdots \mathbf{r}_N) = \sum_{i=1}^N \sum_{j>i}^N (D_{ij} - \langle D \rangle_{ij})^2$$

The value of S will change if one or more of its $N(N-1)/2$ interatomic distances vary but not necessarily when the $3N$ atomic coordinates used to calculate the elements D_{ij} change. Since $\langle \mathbf{D} \rangle$ is invariant to rotation and translation, some restrictions

must be added in S to avoid those changes in $\mathbf{r}_1 \cdots \mathbf{r}_N$ associated with these degrees of freedom. For nonlinear molecules, six terms can be added to S to cause it to increase when rotation and translation occur, with respect to the original geometry^[3]:

$$\begin{aligned}S(\mathbf{r}_1 \cdots \mathbf{r}_N) &= \sum_{i=1}^N \sum_{j>i}^N (D_{ij} - \langle D \rangle_{ij})^2 + \left(\sum_{i=1}^N m_i \Delta x_i \right)^2 \\ &+ \left(\sum_{i=1}^N m_i \Delta y_i \right)^2 + \left(\sum_{i=1}^N m_i \Delta z_i \right)^2 \\ &+ \left(\sum_{i=1}^N m_i (x_i^0 \Delta y_i - y_i^0 \Delta x_i) \right)^2 + \left(\sum_{i=1}^N m_i (x_i^0 \Delta z_i - z_i^0 \Delta x_i) \right)^2 \\ &+ \left(\sum_{i=1}^N m_i (y_i^0 \Delta z_i - z_i^0 \Delta y_i) \right)^2\end{aligned}$$

where $\Delta \alpha_i = \alpha_i - \alpha_i^0$, and $\alpha = x, y$ or z . The coordinates with superscript 0 represent the initial atomic coordinates. In our Fortran implementation of the symmetrization procedure, we use the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm^[29] to search for the coordinates that produce $S(\mathbf{r}_1^{\text{sym}} \cdots \mathbf{r}_N^{\text{sym}}) = 0$. The BFGS method requires the gradient of the function to be minimized. The derivative of S with respect to the x -component of $\mathbf{r}_k$ is given below:

$$\begin{aligned}\left(\frac{\partial S}{\partial x_k} \right) &= 2 \sum_{j \neq k}^N \left[(D_{kj} - \langle D \rangle_{kj}) \frac{(x_k - x_j)}{D_{kj}} \right] + 2m_k \sum_{i=1}^N m_i \Delta x_i \\ &- 2y_k^0 m_k \sum_{i=1}^N m_i (x_i^0 \Delta y_i - y_i^0 \Delta x_i) - 2z_k^0 m_k \sum_{i=1}^N m_i (x_i^0 \Delta z_i - z_i^0 \Delta x_i)\end{aligned}$$

Derivatives with respect to y_k and z_k have similar expressions. An example of symmetrization is shown in Figure 4, where the atoms of the hexanitrocuprate(II) ion were randomly displaced to create an unsymmetrical structure. After symmetrization, the resulting structure has T_h symmetry.

NUMERICAL EXAMPLES

In order to illustrate in detail the complete procedure, the point groups of ammonia and allene will be determined from their Cartesian coordinates and atomic masses. These

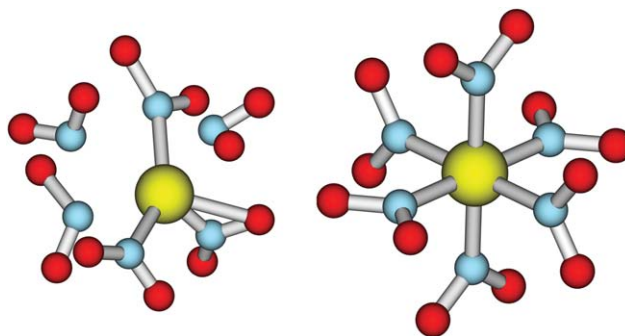


Figure 4. Hexanitrocuprate(II) ion before (left) and after (right) symmetrization. Red spheres represent oxygen and blue nitrogen.

Table 3. Cartesian coordinates and atomic masses of ammonia.

Atom	Mass	Coordinates (in Å)		
	(in <i>u</i>)	<i>x</i>	<i>y</i>	<i>z</i>
N	14.003074	0.000000	0.000000	0.000000
H ₁	1.007825	0.000000	0.000000	1.008000
H ₂	1.007825	0.950353	0.000000	-0.336000
H ₃	1.007825	-0.475176	-0.823029	-0.336000

molecules were randomly oriented to enforce that the presented procedure is independent of the chosen frame.

Ammonia

From the coordinates and atomic masses of ammonia, given in Table 3, the center of mass and inertia tensor are calculated to be $\mathbf{r}_{\text{CM}} = (0.0281, -0.0487, 0.0199)$ and

$$\mathbf{I} = \begin{bmatrix} 1.8871 & -0.4175 & 0.1704 \\ -0.4175 & 2.3692 & -0.2952 \\ 0.1704 & -0.2952 & 1.7666 \end{bmatrix},$$

clearly showing that the Cartesian axes are not the principal axes of inertia. Within the precision of $10^{-4} \text{ Å}^2 \cdot \text{u}^{-1}$, the eigenvalues of this inertia tensor correspond to the oblate symmetric top case ($I_A = I_B = 1.6461$ and $I_C = 2.7307 \text{ Å}^2 \cdot \text{u}^{-1}$). Therefore, this molecule cannot belong to the point groups of

highly symmetric systems and its group will be identified following the procedure of symmetric rotors.

In the next step, the interatomic distance matrix is calculated:

$$\mathbf{D} = \begin{bmatrix} & \text{N} & \text{H}_1 & \text{H}_2 & \text{H}_3 \\ \text{N} & 0.0000 & 1.0080 & 1.0080 & 1.0080 \\ \text{H}_1 & 1.0080 & 0.0000 & 1.6461 & 1.6461 \\ \text{H}_2 & 1.0080 & 1.6461 & 0.0000 & 1.6461 \\ \text{H}_3 & 1.0080 & 1.6461 & 1.6461 & 0.0000 \end{bmatrix}$$

Note that the distances appearing in the column of hydrogen 1 are the same as in the columns of hydrogens 2 and 3, meaning that these atoms are symmetrically equivalent. To determine if this molecule has a proper rotation, the principal moments of the hydrogens are calculated as if they were a single molecule. Their moments are $I_A^H = I_B^H = 1.3654$ and $I_C^H = 2.7307 \text{ Å}^2 \cdot \text{u}^{-1}$. This result shows that these atoms form a regular three-sided polygon because $I_A^H = I_B^H$ and $I_A^H + I_B^H = I_C^H$. Thus, the only possible rotation is the C_3 around the inertia axis of I_C^H whose corresponding eigenvector is $\mathbf{r}^{C_3} = (0.4714, -0.8165, 0.3333)$. The C_3 rotation matrix is given by^[30]

$$\mathbf{C}_3 = \begin{bmatrix} x^2(1-\cos\theta)+\cos\theta & xy(1-\cos\theta)-z\sin\theta & xz(1-\cos\theta)+y\sin\theta \\ xy(1-\cos\theta)+z\sin\theta & y^2(1-\cos\theta)+\cos\theta & yz(1-\cos\theta)-x\sin\theta \\ xz(1-\cos\theta)-y\sin\theta & zy(1-\cos\theta)+x\sin\theta & z^2(1-\cos\theta)+\cos\theta \end{bmatrix}$$

where x , y , and z are the components of $\mathbf{r}^{C_3}$ and $\theta = 120^\circ$. Applying that rotation to the coordinates of ammonia,

$$\mathbf{X} = \begin{bmatrix} \text{N} & \text{H}_1 & \text{H}_2 & \text{H}_3 \\ 0.0000 & 0.0000 & 0.9504 & -0.4752 \\ 0.0000 & 0.0000 & 0.0000 & -0.8230 \\ 0.0000 & 1.0080 & -0.3360 & -0.3360 \end{bmatrix} \begin{matrix} x \\ y \\ x \\ x \end{matrix}$$

the following result is obtained:

$$\mathbf{C}_3 \cdot \mathbf{X} = \begin{bmatrix} \text{N} & \text{H}_1 & \text{H}_2 & \text{H}_3 \\ 0.0000 & -0.4752 & -0.0000 & 0.9504 \\ 0.0000 & -0.8230 & 0.0000 & -0.0000 \\ 0.0000 & -0.3360 & 1.0080 & -0.3360 \end{bmatrix} \begin{matrix} x \\ y \\ y \\ x \end{matrix}$$

The C_3 indeed exists as the only effect from this symmetry operation is the interchange of SEA. The next steps are searching for the $C_2^\perp$, S_6 and S_3 rotations.^[4] In fact, this molecule does not have a $C_2^\perp$ because the center of mass of the symmetrically equivalent hydrogens, $\mathbf{r}_{\text{CM}}^H = (0.1584, -0.2743, 0.1120)$, and of the nitrogen, $\mathbf{r}_{\text{CM}}^N = (0, 0, 0)$, are not the same

point. Checking for S_6 and S_3 is a simple task because these rotations must be collinear to the C_3 axis, $\mathbf{r}^{C_3}$. The improper rotation matrix $\mathbf{S}_6$ is generated by the product $\sigma \cdot \mathbf{C}_6$, where the reflexion plane is given by^[30]

$$\sigma = \begin{bmatrix} 1-2x^2 & -2xy & -2xz \\ -2xy & 1-2y^2 & -2yz \\ -2xz & -2yz & 1-2z^2 \end{bmatrix}$$

with x , y , and z being again the components of $\mathbf{r}^{C_3}$. The resulting coordinates from the action of $\mathbf{S}_6$ are

$$\mathbf{S}_6 \cdot \mathbf{X} = \begin{bmatrix} \text{N} & \text{H}_1 & \text{H}_2 & \text{H}_3 \\ 0.0000 & -0.9504 & 0.4752 & 0.0000 \\ 0.0000 & 0.0000 & 0.8230 & -0.0000 \\ 0.0000 & 0.3360 & 0.3360 & -1.0080 \end{bmatrix} \begin{matrix} x \\ y \\ y \\ x \end{matrix}$$

The situation is different now because none of the new hydrogen positions matches the original one. Therefore, ammonia does not possess the S_6 symmetry operation. The same conclusion holds for the S_3 rotation.

Finally, the existence of reflexion planes must be verified. As described previously any pair of hydrogens, for example, H_1 and H_2 , are chosen to generate a candidate of normal vector:

$$\mathbf{n} = \frac{\mathbf{r}_{H_1} - \mathbf{r}_{H_2}}{\text{norm}(\mathbf{r}_{H_1} - \mathbf{r}_{H_2})}$$

The x , y , and z components of $\mathbf{n}$ are then used to construct the σ matrix as in the improper rotations. Applying this operation to the molecule results in:

$$\sigma \cdot \mathbf{X} = \begin{bmatrix} 0.0000 & 0.9504 & -0.0000 & -0.4752 \\ 0.0000 & 0.0000 & 0.0000 & -0.8230 \\ 0.0000 & -0.3360 & 1.0080 & -0.3360 \end{bmatrix} \begin{matrix} x \\ y \\ x \end{matrix}$$

From this last operation, it can be seen that ammonia does possess the reflection plane which is a σ_v since $\mathbf{n} \cdot \mathbf{r}^{C_3} = 0$. Therefore, one correctly concludes that the ammonia belongs to the C_{3v} point group. As a final remark, note that all symmetry operations were performed with the molecule center of mass out of the origin. In this particular case there was no need of moving the center of mass to the origin because the nitrogen is in the origin.

Allene

The center of mass and the inertia tensor of allene calculated from the coordinates and masses listed in Table 4 are $\mathbf{r}_{CM} = (0.0000, 0.0000, 0.0000)$ and

$$\mathbf{I} = \begin{bmatrix} 42.7012 & -8.5808 & -21.3957 \\ -8.5808 & 51.1936 & -13.2844 \\ -21.3957 & -13.2844 & 23.3974 \end{bmatrix}$$

As in the previous example, the molecule is randomly oriented, that is, the Cartesian axes are not the principal axes of inertia. After diagonalizing $\mathbf{I}$, the principal moments are found to be those of the prolate symmetric top: $I_A = 4.2496$ and $I_B = I_C = 56.5213 \text{ Å}^2 \cdot \text{u}^{-1}$. The procedure to find the point group of this type of rotor is the same as in the previous example. To locate the C_n , first the interatomic distances must be calcu-

Table 4. Cartesian coordinates and atomic masses of allene.

Atom	Mass (in u)	Coordinates (in Å)		
		x	y	z
C_1	12.000000	0.000000	0.000000	0.000000
C_2	12.000000	0.686442	0.426203	1.062720
C_3	12.000000	-0.686442	-0.426203	-1.062720
H_1	1.007825	1.065720	-0.430758	1.617416
H_2	1.007825	0.680464	1.514938	1.085950
H_3	1.007825	-1.732359	-0.558795	-0.789959
H_4	1.007825	-0.013827	-0.525388	-1.913409

lated to separate the atoms into SEA sets. The distance matrix is:

$$\mathbf{D} = \begin{bmatrix} 0.0000 & 1.3350 & 1.3350 & 1.9843 & 1.9843 & 1.9843 & 1.9843 \\ 1.3350 & 0.0000 & 2.6700 & 1.0890 & 1.0890 & 3.2021 & 3.2021 \\ 1.3350 & 2.6700 & 0.0000 & 3.2021 & 3.2021 & 1.0890 & 1.0890 \\ 1.9843 & 1.0890 & 3.2021 & 0.0000 & 2.0534 & 3.6934 & 3.6934 \\ 1.9843 & 1.0890 & 3.2021 & 2.0534 & 0.0000 & 3.6934 & 3.6934 \\ 1.9843 & 3.2021 & 1.0890 & 3.6934 & 3.6934 & 0.0000 & 2.0534 \\ 1.9843 & 3.2021 & 1.0890 & 3.6934 & 3.6934 & 2.0534 & 0.0000 \end{bmatrix}$$

It shows that allene has three sets of SEA: (i) The central carbon, C_1 , (ii) the terminal carbons, and (iii) the four hydrogens. The C_1 carbon is in the molecule center of mass and therefore inadequate for locating symmetry elements. The other two carbons can have any rotations along their axis as well as C_2 rotations perpendicular to this axis. To determine whether the four hydrogens are forming a polygon or a polyhedron, their principal moments are evaluated: $I_A^H = 4.2496$, $I_B^H = I_C^H = 13.7479 \text{ Å}^2 \cdot \text{u}^{-1}$. Their moments fit the prolate symmetric top rotor case meaning that the hydrogens are arranged as a two-sided antiprism. With $k = 4$ the only possible rotation along the prism translation axis is the C_2 . If this C_2 really exists, its axis will be the eigenvector of I_A^H : $\mathbf{r}^{C_2} = (0.5142, 0.3193, 0.7960)$. The initial coordinates,

$$\mathbf{X} = \begin{bmatrix} 0.0000 & 0.6864 & -0.6864 & 1.0657 & 0.6805 & -1.7324 & -0.0138 \\ 0.0000 & 0.4262 & -0.4262 & -0.4308 & 1.5149 & -0.5588 & -0.5254 \\ 0.0000 & 1.0627 & -1.0627 & 1.6174 & 1.0860 & -0.7900 & -1.9134 \end{bmatrix}$$

after the C_2 rotation becomes

$$\mathbf{C}_2 \cdot \mathbf{X} = \begin{bmatrix} 0.0000 & 0.6864 & -0.6864 & 0.6805 & 1.0657 & -0.0138 & -1.7324 \\ 0.0000 & 0.4262 & -0.4262 & 1.5149 & -0.4308 & -0.5254 & -0.5588 \\ 0.0000 & 1.0627 & -1.0627 & 1.0860 & 1.6174 & -1.9134 & -0.7900 \end{bmatrix}$$

confirming the existence of this symmetry operation. The next task is searching for $C_2^\perp$ axes. This molecule can have such rotations because the center of mass of each SEA set, r_{CM}^{SEA} , is the same point: $r_{CM} = r_{CM}^{C_1} = r_{CM}^{C_{2,3}} = r_{CM}^{H_{1-4}} = (0.0000, 0.0000, 0.0000)$. An educated guess for the $C_2^\perp$ axis is obtained from the position of two SEA as explained in the corresponding section. Carbons 2 and 3 cannot be used for this purpose because their middle point $\bar{r}_{C_2,C_3}$ is the center of mass of the molecule. Taking hydrogens 1 and 2 for this task, the first step is computing the middle point vector $\bar{r}_{12}$:

$$\begin{aligned}\bar{r}_{12} &= \frac{r_{H_1} + r_{H_2}}{2} \\ &= (0.8731, 0.5421, 1.3517) \\ &\neq r_{CM}\end{aligned}$$

The condition that $\bar{r}_{12}$ is not equal to r_{CM} enables the next step of evaluating the candidate vector for the $C_2^\perp$ rotation, $r_{C_2^\perp}$:

$$r_{C_2^\perp} = \frac{\bar{r}_{12} - r_{CM}}{\text{norm}(\bar{r}_{12} - r_{CM})}$$

$$= (0.5142, 0.3193, 0.7960)$$

$$= r_{C_2}$$

For the pair H_1 and H_2 , the resulting $r_{C_2^\perp}$ is the r_{C_2} rotation found earlier. Thus, another pair must be tried. Choosing now hydrogens 1 and 3, the middle point will be:

$$\begin{aligned}\bar{r}_{13} &= (-0.3333, -0.4948, 0.4137) \\ &\neq r_{CM}\end{aligned}$$

As earlier we proceed computing vector $r_{C_2^\perp}$:

$$\begin{aligned}r_{C_2^\perp} &= \frac{\bar{r}_{13} - r_{CM}}{\text{norm}(\bar{r}_{13} - r_{CM})} \\ &= (-0.4591, -0.6815, 0.5699) \\ &\neq r_{C_2}\end{aligned}$$

The resulting vector being not the r_{C_2} enables the next step of checking if $r_{C_2^\perp}$ and r_{C_2} are orthogonal: $r_{C_2^\perp} \cdot r_{C_2} = 0$. As they are indeed perpendicular, the last task is the 180° rotation of allene around $r_{C_2^\perp}$. Doing this, one obtains the following coordinates:

$$C_2^\perp \cdot X = \begin{bmatrix} C_1 & C_3 & C_2 & H_3 & H_4 & H_1 & H_2 \\ 0.0000 & -0.6864 & 0.6864 & -1.7324 & -0.0138 & 1.0657 & 0.6805 \\ 0.0000 & -0.4262 & 0.4262 & -0.5588 & -0.5254 & -0.4308 & 1.5149 \\ 0.0000 & -1.0627 & 1.0627 & -0.7900 & -1.9134 & 1.6174 & 1.0860 \end{bmatrix}$$

The resulting coordinates confirm that allene has $C_2^\perp$ symmetry operation. Following the point group flowchart, the existence of the S_4 rotation collinear to C_2 is verified:

$$S_4 \cdot X = \begin{bmatrix} C_1 & C_3 & C_2 & H_4 & H_3 & H_1 & H_2 \\ 0.0000 & -0.6864 & 0.6864 & -0.0138 & -1.7324 & 1.0657 & 0.6805 \\ 0.0000 & -0.4262 & 0.4262 & -0.5254 & -0.5588 & -0.4308 & 1.5149 \\ 0.0000 & -1.0627 & 1.0627 & -1.9134 & -0.7900 & 1.6174 & 1.0860 \end{bmatrix}$$

Because allene possesses the S_4 improper rotation it belongs to the D_{2d} point group, as expected.

Conclusions

A methodology to locate symmetry elements in molecular systems was presented. The search starts with the division of the molecule into subsets of SEA that are analyzed individually. Usually, the number of SEA is lower than the total number of atoms in the molecule; thus, the location of those symmetry

elements in which atoms are taken at random (e.g., reflexion planes) becomes more efficient. Moreover, in those molecules that are either asymmetric or symmetric rotors, the location of rotations is straightforward from the number and inertia tensor of SEA. Conversely, proper rotations are more difficult to locate in spherical rotors but, in these cases, the number of C_2 (found rapidly) can be used to set the molecule point group as T , O or I . From the group information, the number and type of high order rotations (C_3 , C_4 and/or C_5 , depending on the group) are determined, therefore simplifying their location.

The algorithms are easy to implement as the main information required is interatomic distances and inertia tensors of SEA. They are invariant to the molecular orientation and their computational cost is low. We described an algorithm to symmetrize atomic coordinates for slightly unsymmetrical structures, where the new coordinates are found from an averaged interatomic distances matrix. Several examples of outputs are given as Supporting Information. The program is freely available from the authors on request.

Keywords: proper and improper rotation · reflexion plane · point group · SEA · algorithm

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